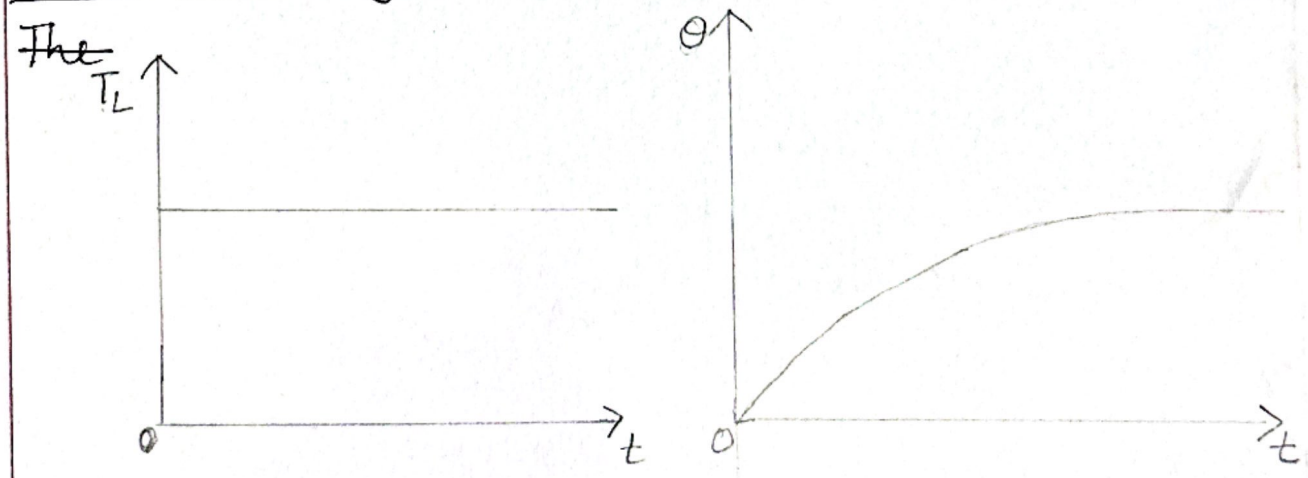
1)a) i) Motor Duty Classes:

The motor duty cycles are divided into eight categories:

1) Continuous duty
2) Short-time duty
3) Intermittent periodic duty
4) Intermittent periodic duty with starting
5) Intermittent periodic duty with starting and braking.
6) Continuous duty with intermittent periodic loading.
7) Continuous duty with starting and braking
8) Continuous duty with periodic speed changes.

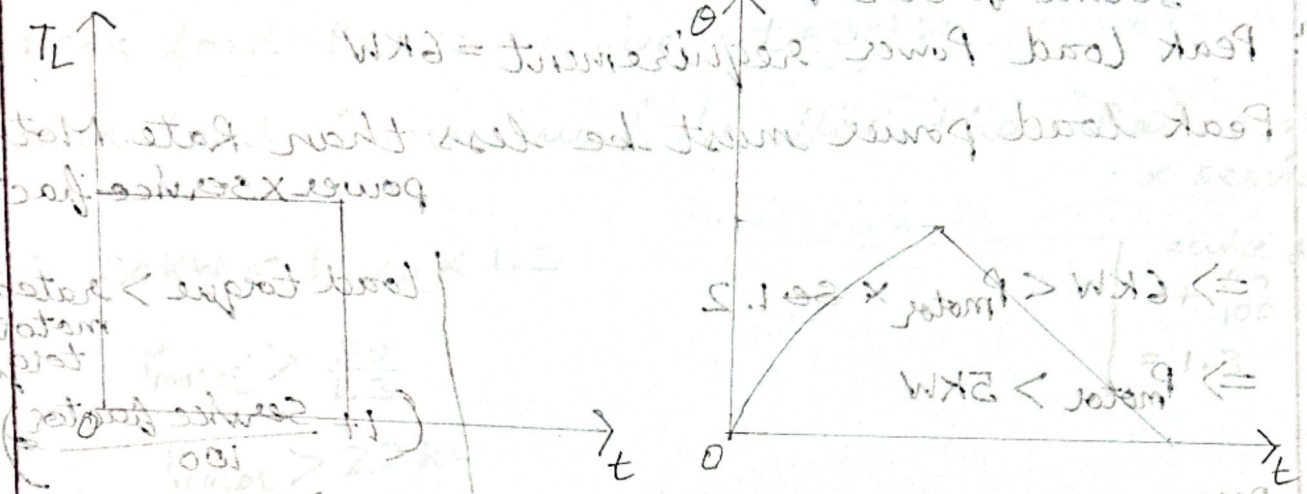
Continuous Duty



The motor runs long enough for its temperature to reach a steady-state. Used in conveyors, compressors.

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Short-Time Duty:



The motor is runs for a short duration, and cools back to ambient temperature before the next run. Used in cranes, household appliances.

ii) Flux change, $\Delta\phi = 2 - 1.4 = 0.6$ Wb

Number of turns, $N = 250$

Time, $\Delta t = 0.45$ s

Induced emf, $E = \frac{N \cdot \Delta\phi}{\Delta t}$

$= \frac{250 \times 0.6}{0.45}$

$= 333.3$ V

1)b). service factor of 30%

Peak load Power requirement = 30KW

Peak load Power must be less than Rate Motor power x service factor

$\Rightarrow$ $30KW < P_{motor} \times 1.3$

$P_{motor} > \frac{30}{1.3}$

$P_{motor} > 23\,KW$

service factor, $1 + \frac{30}{100} = 1.3$

The RMS load power should ideally be between 75% and 100% of Rated Motor Power.

$$\text{RMS load power} = \sqrt{\frac{P_1^2 t_1 + P_2^2 + t_2 + P_3^2 t_3 + \ldots + P_n^2 t_n}{t_1 + t_2 + t_3 + \ldots t_n}}$$

$$= \sqrt{\frac{10^2 \times 9 + 16^2 \times 4 + 20^2 \times 5 + 30^2 \times 20 + 24^2 \times 4}{9+4+5+20+4}}$$

$$= 24.02\,KW$$

If we choose $P_{motor} = 23KW$ then,

$$\text{RMS load power} = \frac{24.02}{23} \times 100$$

$= 104\%$ of P_{motor} which is not acceptable

$\therefore$ choose a bigger motor to meet this requirement.

1)a) An electrical machine is a device that converts mechanical energy to electrical energy (generator) or electrical energy to mechanical energy (motor).

Operation of DC Motor

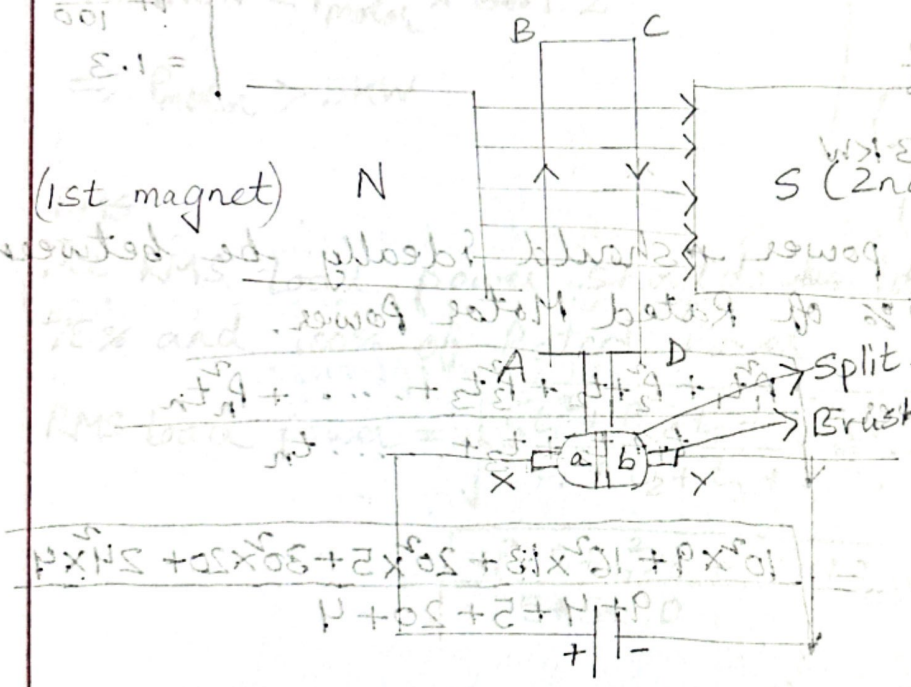


Figure: Electric motor

When current flows in the coil placed between poles, conductor AB experiences downward force and CD upward force (by Fleming's left-Hand Rule), producing rotation. After half a turn, the direction of current is reversed by the split-ring commutator, so the torque continues in the same direction. Thus, the coil keeps rotating continuously until the supply is switched off.

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b) service factor of ~~20%~~ 25%

peak load power required = 6KW

Peak load power must be less than rate motor power × servic factor

$\Rightarrow 6KW < P_{motor} \times 1.25$

$\Rightarrow P_{motor} > 4.8KW$

load torque > rated motor torque

$1 + \frac{25}{100} = 1.25$

The RMS load power should ideally be between 75% and 100% of rated power motor power

$$\text{RMS load power} = \sqrt{\frac{P_1^2 t_1 + P_2^2 t_2 + P_3^2 t_3 + \dots P_n^2 t_n}{t_1 + t_2 + t_3 + \dots t_n}}$$

$$= \sqrt{\frac{4^2 \times 5 + 6^2 \times 5 + 2^2 \times 10}{5+5+10}}$$

$$= 3.87KW$$

If we choose $P_{motor} = 4.8KW$ then,

$$\text{RMS load power} = \frac{3.87}{4.8} \times 100$$

$$= \sout{77.5\%}\ 80.6\%$$

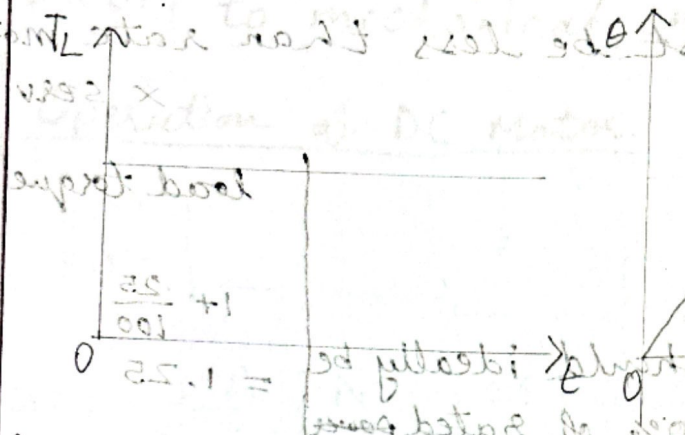
$= 81\%$ of P_{motor} which is acceptable

∴ Choose a 4.8KW motor

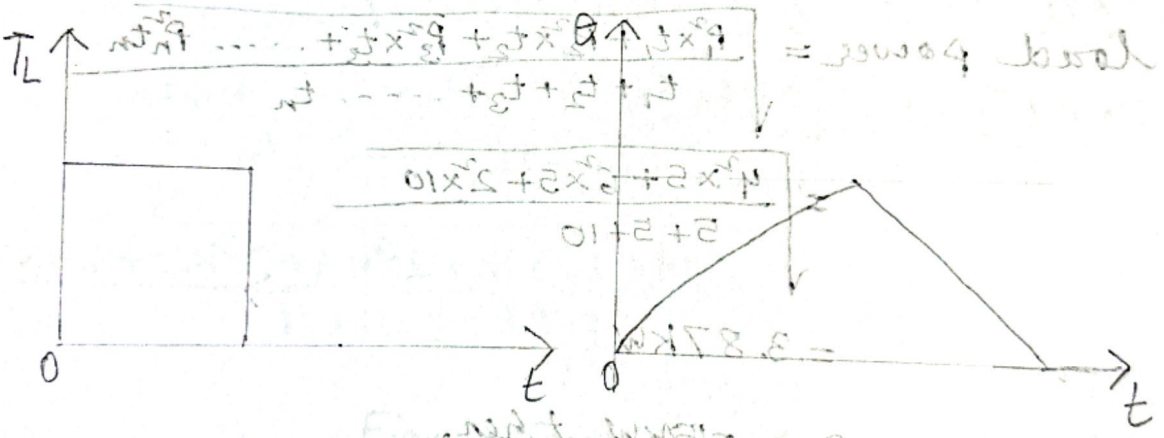
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c) Motor Duty Classes

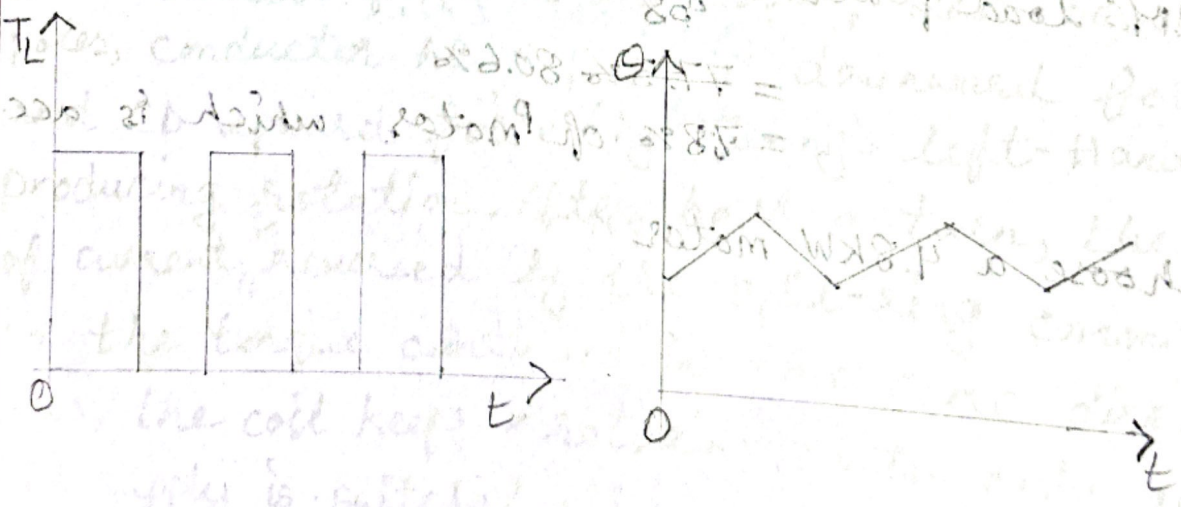
1) Continuous Duty



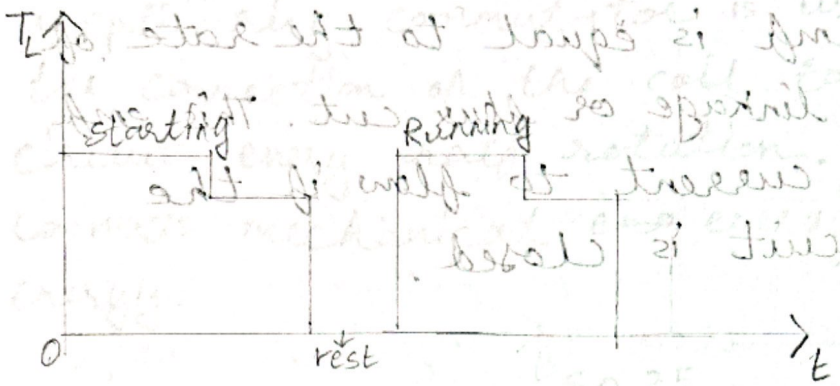
2) Short Time Duty



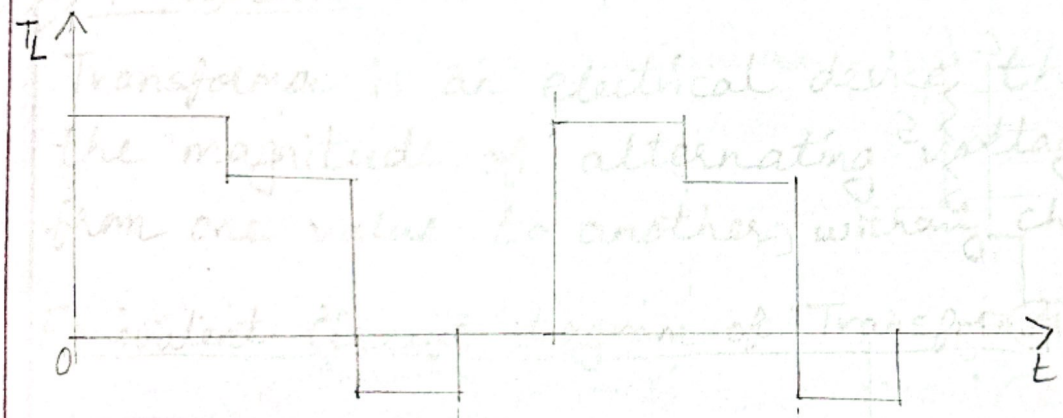
3) Intermittent Periodic Duty

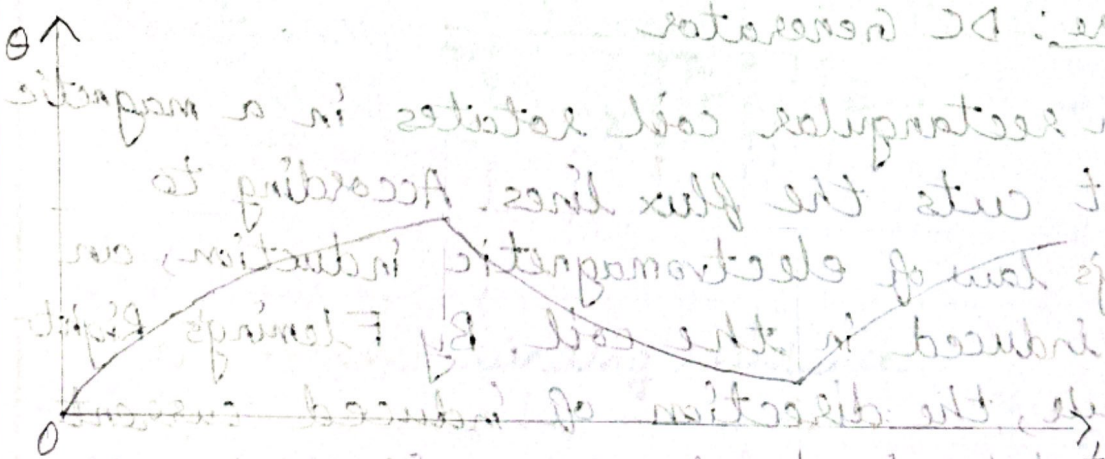


4) Intermittent Periodic Duty with Starting



5) Intermittent Periodic Duty with Starting & Breaking





1)a) Faraday's Second law:

The induced emf is equal to the rate of change of flux linkage or flux cut. This emf will cause a current to flow if the conductor circuit is closed.

~~b)~~

c) Operation of DC Generator

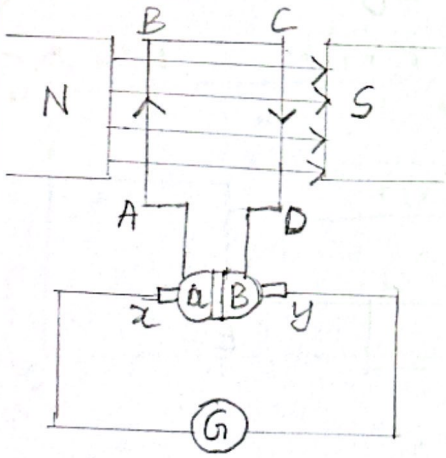


Figure: DC Generator

When a rectangular coil rotates in a magnetic field, it cuts the flux lines. According to Faraday's law of electromagnetic induction, an emf is induced in the coil. By Fleming's Right Hand rule, the direction of induced current can be determined. As the coil rotates, the direction of current in its sides reverses every

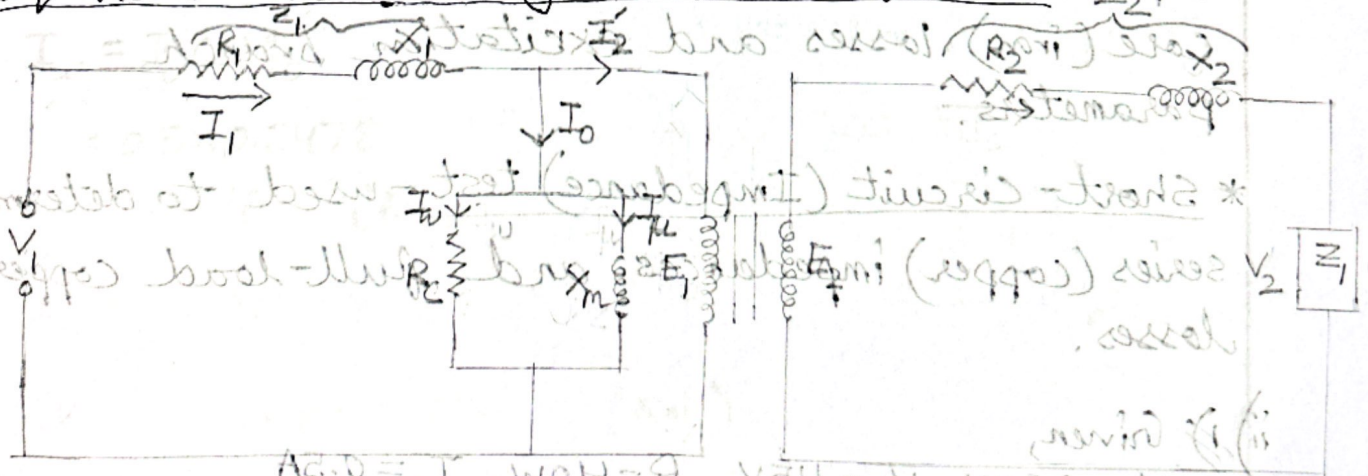
half turn. To obtain unidirectional current, a split-ring commutator is used, which reverses the connection of the coil to the external circuit every half rotation. Thus, a DC generator converts mechanical ~~ene~~ energy into direct electrical energy.

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2)a)i) Transformer

Transformer is an electical device that changes the magnitude of alternating voltage or current from one value to another, without changing the frequency

Equivalent Circuit diagram of Transformer



Here, R_1 = Primary winding resistance
R_2 = Secondary "
I_μ = Magnetizing component
I_w = Working component
I_0 = No load current

Z = impedance
X = reactance

ii) Core Losses in Transformer

Core (iron) losses occur in the magnetic core due to alternating flux. They are:

1) Hysteresis loss - due to repeated magnetization and demagnetization, proportional to frequency and area of hysteresis loop.

2) Eddy Current loss - induced circulating currents in the core, producing heat; minimized by ~~laminated~~ laminating the core.

2)b) i) ~~Two~~ Transformer Tests

* Open-circuit (No-load) test - used to determine core (iron) losses and excitation branch parameters.

* Short-circuit (Impedance) test - used to determine series (copper) impedances and full-load copper losses.

ii) Given,

$V_1 = 230V$, $V_2 = 115V$, $P = 40W$, $I_0 = 0.5A$

1) Turn ratio (Primary : secondary)

$$\frac{N_1}{N_2} \approx \frac{V_1}{V_2} = \frac{230}{115} = 2$$

i.e. 2 : 1 (Ans)

$P = V_1 I_0 \cos\phi_0$

$I_\mu = I_0 \sin\phi_0$

$I_w = I_0 \cos\phi_0$

2) No-load power factor / $\cos\phi_0$

$$P.F = \cos\phi_0 = \frac{P}{V_1 I_0}$$

$$= \frac{40}{230 \times 0.5} = 0.3478 \quad \text{(Ans)}$$

~~Magnetizing Component~~

$\sin\phi_0 = \sqrt{1 - \cos^2\phi_0}$

~~$I_\mu = I_0 \sin\phi$~~

2) Magnetizing component, I_μ

$$\sin\phi_0 = \sqrt{1 - \cos^2\phi_0}$$

$$= \sqrt{1 - (0.3478)^2}$$

$$= 0.9376$$

$$I_\mu = I_0 \sin\phi_0$$

$$= 0.5 \times 0.9376$$

$$= 0.469A \quad \text{(Ans)}$$

3) Working component, I_w

$$I_w = I_0 \cos\phi_0$$

$$= 0.5 \times 0.3478$$

$$= 0.174A \quad \text{(Ans)}$$

4) Iron loss

Iron loss = $W_i P = 40W$ (Ans)

2b) or)

i) Distinguish between the transformer tests:

* Open-circuit (No-load) test: Applied to the low-voltage winding with the high-voltage winding open. Determines iron (core) loss and excitation-branch parameters R_0 and X_0

* Short-circuit (Impedance) test: One winding is shorted and a reduced voltage applied to the other. Determines series (copper) impedance and full-load copper loss; excitation branch is negligible during this test.

ii) Given: transformer rating

5kVA, 200/1000 V, 50 Hz

Open-circuit test on L.V. side:

$V_{oc} = 200\,V$, $I_{oc} = 1.2\,A$, $P_{oc} = 90\,W$

Short-circuit test on H.V. side:

$V_{sc} = 50\,V$, $I_{sc} = 5\,A$, $P_{sc} = 110\,W$

iii) Turns ratio:

$$a = \frac{V_{LV}}{V_{HV}} = \frac{200}{1000} = 0.2$$

2)

No-load power factor:

$$\cos\phi_0 = \frac{P_{oc}}{V_{oc} \cdot I_{oc}} = \frac{90}{200 \times 1.2} = 0.375$$

Magnetizing & working components:

$$I_w = I_{oc}\cos\phi_0 = 1.2 \times 0.375 = 0.45A$$

$$I_\mu = I_{oc}\sin\phi_0$$

$$= I_{oc}\sqrt{1-\cos^2\phi_0}$$

$$= I_0\, 1.2\sqrt{1-(0.375)^2}$$

$$= 1.1124A$$

Excitation branch impedances:

$$R_0 = \frac{V_{oc}}{I_w} = \frac{200}{0.45} = 444.44\Omega$$

$$X_0 = \frac{V_{oc}}{I_\mu} = \frac{200}{1.1124} = 179.79\Omega$$

Measured short-circuit impedance on HV side:

$$Z_{sc} = \frac{V_{sc}}{I_{sc}} = \frac{50}{5} = 10\Omega$$

Series resistance on HV side from measured loss:

$$R_{eq} = \frac{P_{sc}}{I_{sc}^2}$$

$$= \frac{110}{5^2} = 4.4\Omega$$

Impedance formula

$$Z^2 = R^2 + X^2$$

$$\angle Z = \tan^{-1}\left(\frac{X}{R}\right)$$

Refer series impedance to L.V side using

~~$a^2 = 0.2$~~

$$Z_{eq} = a^2 Z_{sc} = (0.2)^2 \times 10 = 0.4\,\Omega$$

~~$R_{eq} = a^2 R_{eq}$~~

$$R_{eq(LV)} = a^2 R_{eq(HV)} = 0.2^2 \times 4.4 = 0.176\,\Omega$$

$$X_{eq} = \sqrt{Z_{eq}^2 - R_{eq}^2} = \sqrt{(0.4)^2 - (0.176)^2} = 0.359\,\Omega$$

Sp 24

2)a) Transformer rating is given in VA (KVA, MVA) instead of KW because

* Copper loss depends on current.
* Core loss depends on voltage.
* Since losses are independent of load power factor, transformer rating is expressed in VA, not KW.

b) ~~N_1~~ = Given:

$N_1 = 400$, $N_2 = 1000$, $A = 60\,cm^2 = 60 \times 10^{-4}\,m^2$

$f = 50\,Hz$, $V_1 = 520\,V$ (rms).

i) For a sinusoidal emf:

$$E_1 = 4.44 f N_1 \Phi_m$$

$$\Phi_m = \frac{E_1}{4.44 f N_1} = \frac{520}{4.44 \times 50 \times 400} = 5.85 \times 10^{-3}\,Wb$$

Flux density:

$$B_m = \frac{\phi_m}{A} = \frac{5.85\times10^{-3}}{60\times10^{-4}} = 0.975\,T$$

$B = \Phi/A$

B: magnetic flux density

~~Φ: flux density~~

~~Φ~~ $= \phi$: magnetic flux

ii) $E_2 = 4.44 f N_2 \phi_m$

$= 4.44\times50\times1000\times5.85\times10^{-3}$

$= 1298\,V$

~~2c)~~ 08) Rule for referring impedances (resistance or reactance)

* From LV to HV side: multiply by a^2
* From HV to LV " : divide by a^2

2c) OR) Given:

50kVA, 4400/220V $\Rightarrow a = \frac{V_{HV}}{V_{LV}} = \frac{4400}{220} = 20$

$R_1 = 3.45\,\Omega$, $X_1 = 5.20\,\Omega$ (HV winding)

$R_2 = 0.009\,\Omega$, $X_2 = 0.015\,\Omega$ (LV winding)

i) Parameters referred to HV side

$R_2' = a^2 \times R_2 = 20^2 \times 0.009 = 3.6\,\Omega$

$X_2' = a^2 \times X_2 = 20^2 \times 0.015 = 6\,\Omega$

Now add series terms:

$R_{eq(HV)} = R_1 + R_2' = 3.45 + 3.6 = 7.05\,\Omega$

$X_{eq(HV)} = X_1 + X_2' = 5.20 + 6 = 11.2\,\Omega$

$Z_{eq(HV)} = \sqrt{R_{eq}^2 + X_{eq}^2} = \sqrt{(7.05)^2 + (11.2)^2} = 13.23\,\Omega$

$\angle Z = \tan^{-1}\left(\frac{X_{eq}}{R_{eq}}\right) = \tan^{-1}\left(\frac{11.2}{7.05}\right) = 57.8°$

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ii) Parameters referred to LV side

$$R_1' = a^2 \frac{R_1}{a^2} = \frac{3.45}{20^2} = 0.008625\,\Omega$$

$$X_1' = \frac{X_1}{a^2} = \frac{5.20}{20^2} = 0.013\,\Omega$$

Now add series terms:

$$R_{eq(LV)} = R_1' + R_2 = 0.008625 + 0.009 = 0.017625$$

$$X_{eq(LV)} = X_1' + X_2 = 0.013 + 0.015 = 0.028\,\Omega$$

$$Z_{eq(LV)} = \sqrt{R_{eq}^2 + X_{eq}^2} = \sqrt{(0.017625)^2 + (0.028)^2}$$

$$= 0.03308\,\Omega$$

$$\angle Z = \tan^{-1}\left(\frac{X_{eq}}{R_{eq}}\right) = \tan^{-1}\left(\frac{0.028}{0.017625}\right)$$

$$= 57.8^\circ$$

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2b) Given:

at 1000 V, 50 Hz

$P_{C_1} = 1000\,W$, $P_{h_1} = 650\,W$, $P_{e_1} = 350\,W$

Hysteresis loss $\propto f \times V^{1.6}$

$$P_{h_2} = P_{h_1}\left(\frac{V_2}{V_1}\right)^{1.6} \times \left(\frac{f_2}{f_1}\right)$$

$$= 650 \times \left(\frac{2000}{1000}\right)^{1.6} \times \left(\frac{100}{50}\right)$$

$$= 3940\,W$$

$V_1 = 1000\,V$
$V_2 = 2000\,V$
$f_1 = 50\,Hz$
$f_2 = 100\,Hz$

Eddy loss $\propto f^2 \times V^2$

$$P_{e_2} = P_{e_1} \times \left(\frac{f_2}{f_1}\right)^2 \times \left(\frac{V_2}{V_1}\right)^2$$

$$= 350 \times \left(\frac{100}{50}\right)^2 \times \left(\frac{2000}{1000}\right)^2$$

$$= 5600\,W$$

Total core loss,

$$P_{C_2} = P_{h_2} + P_{e_2}$$

$$= 3940 + 5600$$

$$= 9540\,W$$

c) Given:

$V_1 = 220$, $V_2 = 110V$, $I_0 = 0.5A$, $P = 30W$

i) Turn ratio

$$\frac{N_1}{N_2} \approx \frac{V_1}{V_2} = \frac{220}{110} = 2$$

i.e 2:1

ii) Power factor,

$$\cos\phi_0 = \frac{P}{I_0 V_1} = \frac{30}{0.5 \times 220}$$

$$= 0.2727$$

$P = I_0 V_1 \cos\phi_0$

Magnetizing Component,

$$I_\mu = I_0 \sin\phi_0$$

$$= I_0 \sqrt{1 - \cos^2\phi_0}$$

$$= 0.5\sqrt{1 - (0.2727)^2} = 0.481A$$

iii) Working component,

$I_W = I_0 \cos\phi_0$

$= 0.5 \times 0.2727$

$= 0.1364A$

iv) Iron loss, $= P = 30W$

OR

2)

2)a) Working Principle:

* A transformer works on Faraday's law of electromagnetic induction
* An alternating voltage applied to the primary winding produces an alternating magnetic flux in the laminated iron core.
* This flux links both windings and induces emfs E_1 and E_2 in the primary and secondary according to Faraday's law.
* If the secondary winding is connected to a load, the induced emf causes current to flow and power is transferred from primary to secondary without electrical contacts, only through mutual induction

b) Transformer tests (open-circuit & short-circuit) are performed to determine:

* Core (iron) losses and copper losses without loading the transformer.
* Equivalent circuit parameters (resistances & reactances)
* Efficiency and voltage regulation at different loads

Difference between Ideal and Practical transformer

Ideal transformer	Practical transformer
1) There is no winding resistance	1) Winding resistance is present
2) There is no leakage flux	2) Leakage flux is present
3) There is no core losses	3) Core losses is present
4) 100% efficient	4) Efficiency always less than 100%

3)a) ii) Laws and Rules:

*Faraday's First Law: Whenever a conductor rotates or cuts magnetic flux, an EMF is induced in it.

*Faraday's second Law: The induced EMF is proportional to the rate of change of flux linkage. $E = -N\frac{d\Phi}{dt}$

*Flemings Right-Hand Rule: Thumb → motion of conductor, Index finger → direction of magnetic field, Middle finger → direction of induced current.

*Flemings Left-Hand Rule: Thumbs → motion on conductor, Index finger → magnetic field, Middle finger → direction of current.

b) ii) Given:

Load current, $I_L = 455A$

Terminal voltage, $V = 230V$

Shunt field resistance, $R_{sh} = 40\Omega$

Armature resistance, $R_a = 0.03\Omega$

Shunt field current,

$$I_{sh} = \frac{V}{R_{sh}} =$$

$$= \frac{230}{40} = 5.75\,A$$

Armature current,

$$I_a = I_L + I_{sh} = 455 + 5.75 = 460.75A$$

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Generated EMF,

$E_g = V + I_a R_a$

$= 230 + (460.75 \times 0.03)$

$= 243.82\,V$

$= 244\,V$

OR

3)b)ii) Given:

Terminal voltage, $V = 255\,V$

No-load speed, $n_o = 1000\,rpm$,

No-load line current, $I_{o(total)} = 9A$

Field resistance, $R_f = 255\,\Omega$

Field current, $I_f = \frac{V}{R_f} = \frac{255}{255} = 1A$

Armature resistance, $R_a = 0.2\,\Omega$

Loaded total line current, $I_{L(total)} = 50A$

~~Armature current~~

No-load armature current,

$I_{ao} = I_{o(total)} - I_f$

$= 9 - 1$

$= 8A$

Loaded armature current,

$I_{aL} = I_{L(total)} - I_f$

$= 50 - 1$

$= 49A$

No-load back EMFs,

$E_o = V - I_{ao} R_a = 255 - 8(0.2) = 253.4\,V$

Loaded back EMF,

$E_L = V - I_{aL} R_a = 255 - 49(0.2)$

$= 245.2V$

Speed, $n_L = n_0 \times \frac{E_L}{E_0}$

$= 1000 \times \frac{245.2}{253.4}$

$= 967.6 \text{ rpm}$

$= 968 \text{ rpm}$

Sp 24

3)b) Given:

Number of turns, $N = 700$

Induced EMF, $E = 50V$

Time interval, $t = 0.7s$

$E = N \frac{\Delta\phi}{\Delta t}$

$\Delta\phi = \frac{E \times \Delta t}{N}$

$= \frac{50 \times 0.7}{700}$

$= 0.05 Wb$

c) Given:

Load current, $I_L = 450A$

Terminal voltage, $V = 230V$

Shunt field resistance, $R_{sh} = 50\Omega$

Armature resistance, $R_a = 0.03\Omega$

shunt field current,

$$I_{sh} = \frac{V}{R_{sh}} = \frac{230}{50}$$
$$= 4.6A$$

Armature current,

$$I_a = I_L + I_{sh}$$
$$= 450 + 4.6$$
$$= 454.6A$$

Generated EMF,

$$E_g = V + I_a R_a$$
$$= 230 + (454.6 \times 0.03)$$
$$= 243.64V$$
$$= 244V$$

Aus:23

3) Given:

30kVA, 2400/120 V $\Rightarrow a = \frac{V_{HV}}{V_{LV}} = \frac{2400}{120} = 20$

$R_1 = 0.10\Omega$, $X_1 = 0.22\Omega$ (HV)

$R_2 = 0.035\Omega$, $X_2 = 0.012\Omega$ (LV)

i) Referred to HV side

$$R_2' = a^2 R_2 = 20^2 \times 0.035 = 14\Omega$$

$$X_2' = a^2 R a^2 X_2 = 20^2 \times 0.012 = 4.8\Omega$$

Series sums:

$$R_{eq(HV)} = R_1 + R_2' = 0.10 + 14 = 14.10\Omega$$

$$X_{eq(HV)} = X_1 + X_2' = 0.22 + 4.8 = 5.02\Omega$$

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$$Z_{eq(HV)} = \sqrt{R_{eq}^2 + X_{eq}^2}$$

$$= \sqrt{14.10^2 + 5.02^2}$$

$$= 14.97\,\Omega$$

$$\angle Z = \tan^{-1}\left(\frac{X_{eq}}{R_{eq}}\right)$$

$$= \tan^{-1}\left(\frac{5.02}{14.10}\right)$$

$$= 19.6^\circ$$

ii) Referred to LV side

$$R_1' = \frac{R_1}{a^2} = \frac{0.10}{20^2} = 0.00025\,\Omega$$

$$X_1' = \frac{X_1}{a^2} = \frac{0.22}{20^2} = 0.00055\,\Omega$$

Series sums:

$$R_{eq(LV)} = R_1' + R_2 = 0.00025 + 0.035 = 0.03525\,\Omega$$

$$X_{eq(LV)} = X_1' + X_2 = 0.00055 + 0.012 = 0.01255\,\Omega$$

$$Z_{eq(LV)} = \sqrt{R_{eq}^2 + X_{eq}^2}$$

$$= \sqrt{(0.03525)^2 + (0.01255)^2}$$

$$= 0.0374\,\Omega$$

$$\angle Z = \tan^{-1}\left(\frac{X_{eq}}{R_{eq}}\right)$$

$$= \tan^{-1}\left(\frac{0.01255}{0.03525}\right)$$

$$= 19.6^\circ$$